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Advanced Drone GIS Processing Pipeline

This report summarizes the automated pipeline engineered to process uncompressed drone telemetry and high-resolution aerial imagery into professional-grade geographic information system (GIS) assets.

Computer Vision & Seamless Mosaicking

The pipeline utilizes OpenCV and Scale-Invariant Feature Transform (SIFT) algorithms to detect thousands of unique keypoints across the dataset. These points undergo RANSAC homography estimation, successfully aligning and warping the 52 standalone images into a singular, seamless panoramic mosaic. Lighting discrepancies are minimized using CLAHE equalization.

Georeferencing & Orthomosaic Generation

Extracting precise GPS coordinates and altitude data from the DJI EXIF metadata, the system calculates the exact spatial boundaries. Utilizing the Rasterio library, the panoramic mosaic is accurately projected onto a geographic grid (EPSG:4326), generating an industry-standard GeoTIFF orthomosaic ready for integration into Earth observation platforms.

Structure from Motion (SfM) 3D Reconstruction

By evaluating the parallax and parallax shift between overlapping 2D images, the pipeline reconstructs the original camera poses. It then triangulates the depth of matched features to generate a sparse 3D point cloud, allowing for interactive volumetric visualization of the target terrain.

Vegetation Analysis (NCVI)

To assess agricultural health without specialized infrared sensors, the system computes the Natural Color Vegetation Index (NCVI). By comparing green and red reflectance bands, the algorithm generates a diagnostic heatmap that highlights zones of dense, healthy vegetation versus barren soil and infrastructure.

Conclusion

The resulting assets—the georeferenced orthomosaic, 3D SfM point cloud, and NCVI analytics—provide a highly accessible, rapid-deployment solution for precision agriculture and environmental monitoring.